THE FIRST EXT OF A QUOTIENT BY A FINITELY GENERATED IDEAL FITS INTO A CANONICAL SHORT EXACT SEQUENCE WITH THE KOSZUL HOMOLOGY AND THE HOM OF THE NAIVE KOSZUL MODULE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement span contains the author's displayed definition of H_1(N, f_bullet), whose denominator is written with an unbalanced parenthesis, namely '{f_1x, \ldots, f_nx) \mid x \in N}'; that is the author's text and is reproduced unchanged. The four references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The six-term exact sequence for Ext attached to a short exact sequence, and Ext_R(R, N) = 0. invoked at proof lines 24433-24436. Gives the description of Ext_R(R/I, N) as a cokernel.
- The short exact sequence of the naive Koszul complex and its module K. invoked at proof lines 24441-24445. Produces the three-term complex whose middle homology is the Ext group.
- The explicit description of the Koszul homology H_1(N, f_bullet) as a quotient of relations by the Koszul boundary. used at proof lines 24416-24419 and 24448-24450. Identifies the cokernel of the first map of the complex with the module in the statement.

Source reference. Original lemma 05E2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6975–6975. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 010I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1328–1328. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 24394: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24394–24394. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let $I = (f_1, \ldots, f_n)$ be a finitely generated ideal of R. For any R-module N set*

$$H_1(N, f_\bullet) = \frac{\{(x_1, \ldots, x_n) \in N^{\oplus n} \mid f_i x_j = f_j x_i\}}{\{f_1 x, \ldots, f_n x) \mid x \in N\}}$$

For any R-module N there exists a canonical short exact sequence

$$0 \to \mathrm{Ext}_R(R/I, N) \to H_1(N, f_\bullet) \to \mathrm{Hom}_R(K, N)$$

where K is as in Lemma 24394.

Proof. The notation above indicates the Ext-groups in Mod_R as defined in Homology, Section 010I. These are denoted $\mathrm{Ext}_R(M, N)$. Using the long exact sequence of Homology, Lemma 05E2 associated to the short exact sequence $0 \to I \to R \to R/I \to 0$ and the fact that $\mathrm{Ext}_R(R, N) = 0$ we see that

$$\mathrm{Ext}_R(R/I, N) = \mathrm{Coker}(N \longrightarrow \mathrm{Hom}(I, N))$$

Using the short exact sequence of Lemma 24394 we see that we get a complex

$$N \to \mathrm{Hom}(M, N) \to \mathrm{Hom}_R(K, N)$$

whose homology in the middle is canonically isomorphic to $\mathrm{Ext}_R(R/I, N)$. The proof of the lemma is now complete as the cokernel of the first map is canonically isomorphic to $H_1(N, f_\bullet)$. □